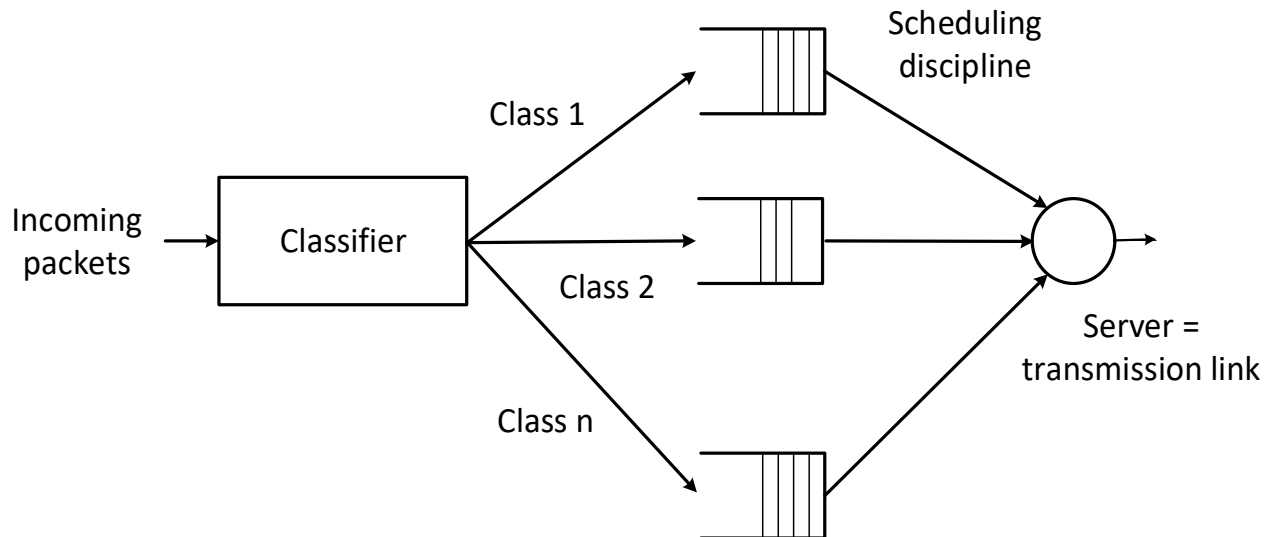


Packet scheduling

Packet scheduling

- Packet scheduling discipline – decides the order of packet transmissions at a link
- An important mechanism for QoS differentiation
- Incoming packets are classified into classes and transmitted according to some scheduling discipline

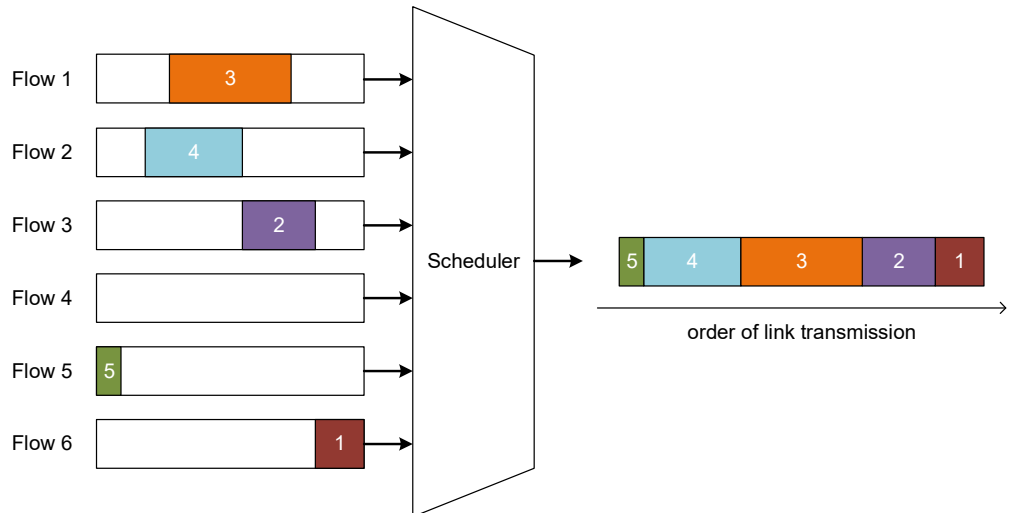


Packet scheduling

- Two types of disciplines:
 - With work conservation – the transmission link is only inactive when no packets are waiting to be transmitted
 - Without work conservation – the transmission link can be inactive while packets are waiting to be transmitted
- Work conserving disciplines
 - By order of arrival: FIFO
 - Based on strict priorities
 - Round robin: Weighted Round Robin, Deficit Round Robin
 - Approximating the GPS system: Weighted Fair Queuing
 - ...
- Non-work conserving disciplines
 - Rate-controlled strict priority
 - ...

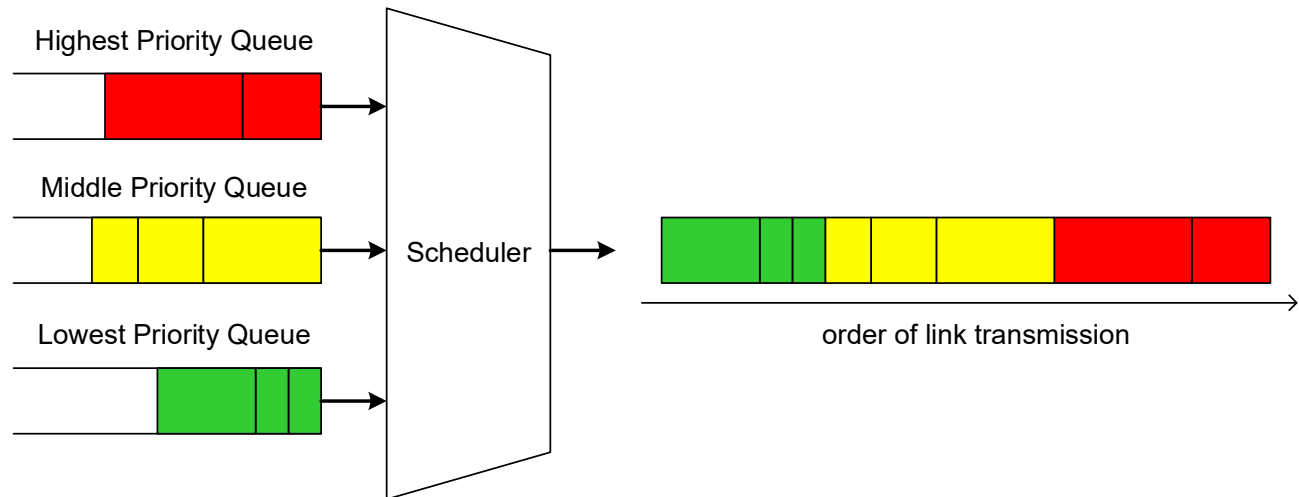
First-In First-Out (FIFO)

- Also known^{as} First-Come First-Served (FCFS)
- Packets are served by order of arrival
- Default packet scheduling mechanism used in IP routers
- Service given to specific packet flows can not be differentiated
- No protection against misbehaved flows



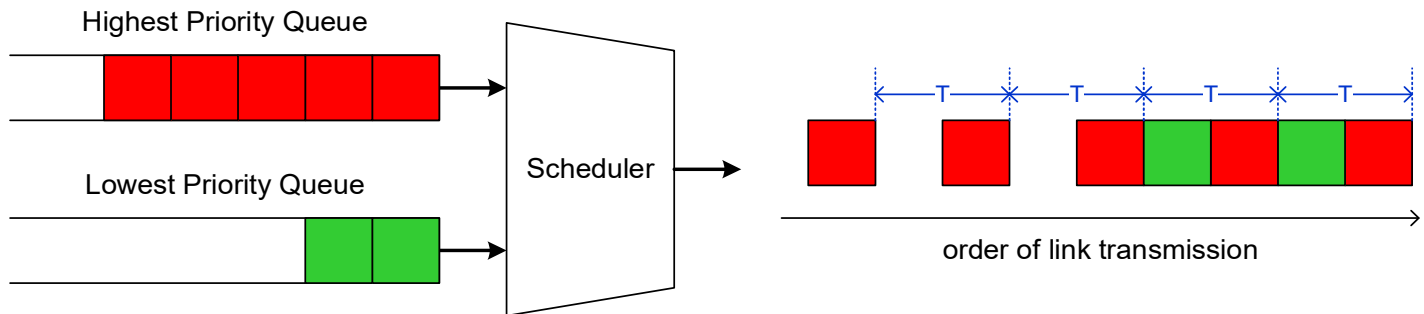
Strict priority

- Each class is assigned a priority
- Packets with higher priority are always transmitted before packets with lower priority
- A higher priority packet flow can degrade the service of lower priority ones (starvation)



Rate-controlled strict priority

- Strict priority scheduling but queues are constrained to not exceed a certain percentage of the link capacity
- Is this scheme work-conserving?

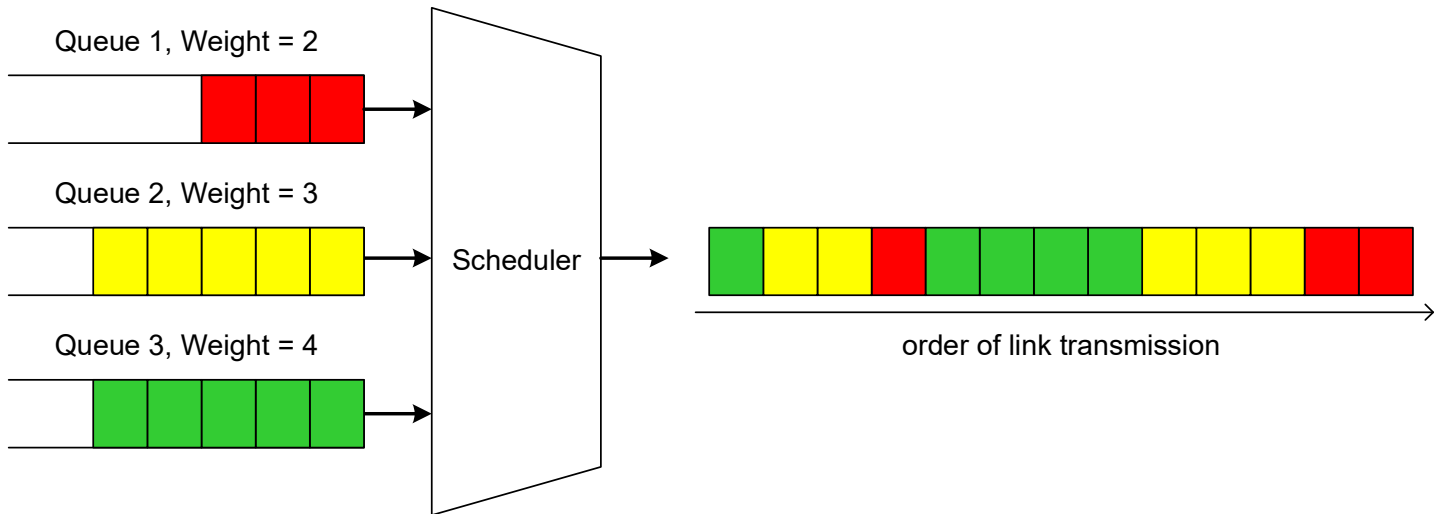


Time interval between successive transmissions of each queue not smaller than $T \rightarrow$ Transmission rate not larger than $1/T$ packets/s

Weighted Round-Robin

- Each class is assigned a queue and each queue is assigned a weight
- The queues are served in round-robin order
- The queue weight defines the guaranteed minimum percentage of the link capacity that a queue will get
- Fixed size packets:
 - Every time a queue is visited, transmit a number of packets up to the queue weight before proceeding to the next queue
 - If queue gets empty, proceed immediately to the next queue
- Variable packet lengths:
 - Same procedure but needs to know a priori the average packet length of each flow in order to calculate the correct queue weights
- Can stay for too long serving a single flow

Weighted Round-Robin (fixed-size packets)

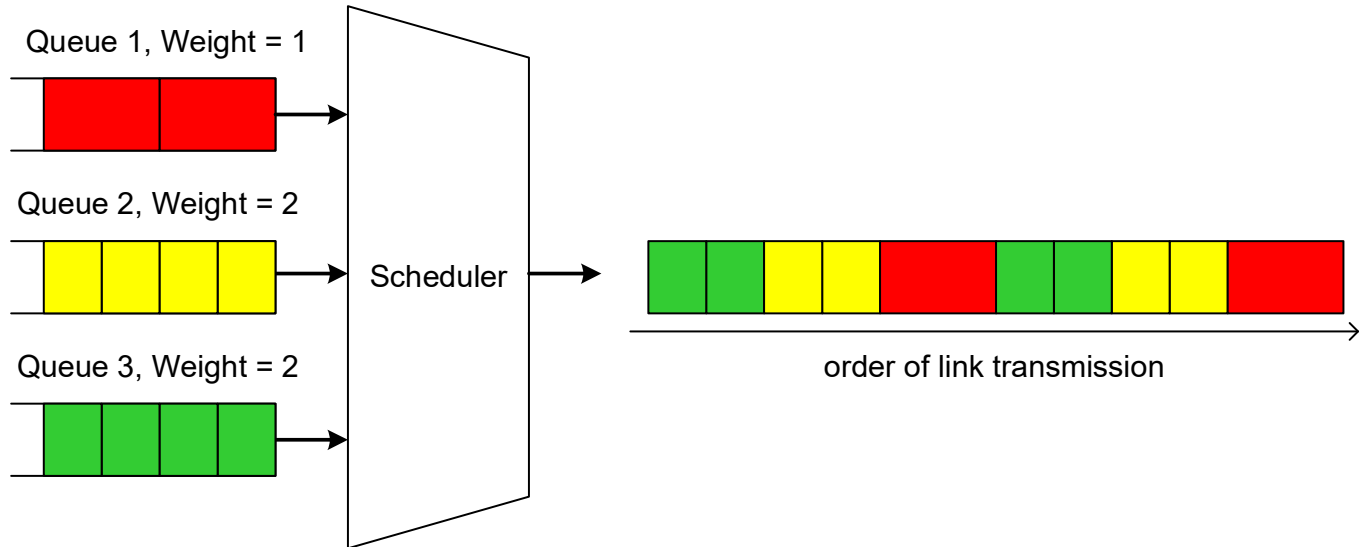


If link has capacity C bits/s, the minimum guaranteed rate of queue i , with weight w_i , is:

$$r_i = \frac{w_i}{\sum_i w_i} C$$

Maximum achievable rate (any queue): C bits/s

Weighted Round-Robin (variable-sized packets)



- Problem with variable-sized packets: all queues get the same percentage of the link capacity (in bits/second), although the weights of queues 2 and 3 are the double of queue 1
- Can determine appropriate weights, but needs to know average packet size associated to each queue

Deficit Round-Robin

- Handles variable packets sizes without knowing the average packet size of each flow in advance
- Each class is assigned a queue and each queue is assigned a quantum (in bytes); the queues are served in round-robin order; the quantum defines the guaranteed minimum percentage of the link capacity that a queue will get
- Operation:
 - credits for transmission*
 - Every queue has a deficit counter initialized to 0
 - Every time a visit to a queue is initiated add quantum bytes to the deficit counter; if the queue is found empty reset the deficit counter to 0
 - While visiting a queue one or more packets can be transmitted; a packet can be transmitted if its packet size is equal or lower to the deficit counter; when a packet is transmitted subtract its size from the deficit counter; if no more packets can be transmitted proceed to the next queue immediately
- Can stay for too long serving a single flow

Deficit Round-Robin

If link has capacity C bits/s, the minimum guaranteed rate of queue i , with quantum q_i , is:

$$r_i = \frac{q_i}{\sum_i q_i} C$$

Maximum achievable rate (any queue): C bits/s

1st cycle:

- Q1 not served, credit Q1 = 1000
- P1-Q2, credit Q2 = 2000 - 1500 = 500
- P2-Q2, credit Q2 = 500 - 200 = 300
- P3-Q2, credit Q2 = 300 - 100 = 200

2nd cycle:

- P1-Q1, credit Q1 = 2000 - 1300 = 700
- P2-Q1, credit Q1 = 700 - 500 = 200
- P4-Q2, credit Q2 = 2200 - 900 = 1300

3rd cycle:

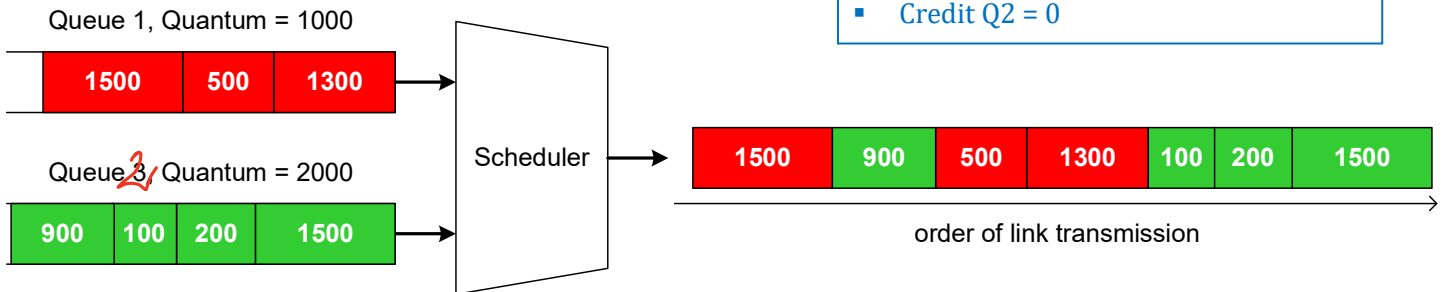
- Q1 not served, credit Q1 = 1200
- Credit Q2 = 0

4th cycle:

- P3-Q1, credit Q1 = 2200 - 1500 = 700
- Credit Q2 = 0

5th cycle:

- Credit Q1 = 0
- Credit Q2 = 0



Weighted Fair Queuing

- Each class is assigned a queue and each queue is assigned a weight; weight defines the guaranteed minimum percentage of the link capacity that a queue will get
- Like WRR and DRR aims at providing minimum rate guarantees. But it tries to approach the ideal situation of GPS (Generalized Processor-Sharing) systems
- Packets are transmitted by the order of finish times in the corresponding GPS system

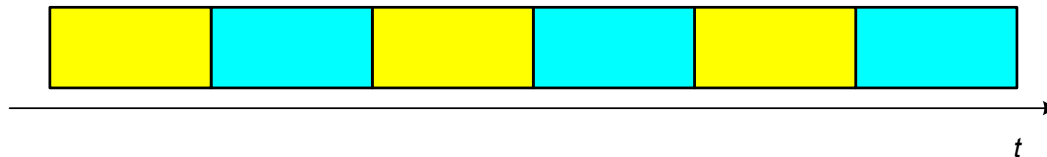
Generalized Processor Sharing (GPS)

- Ideal system
 - Traffic is infinitely divisible
 - The “link” can transmit simultaneously multiple flows

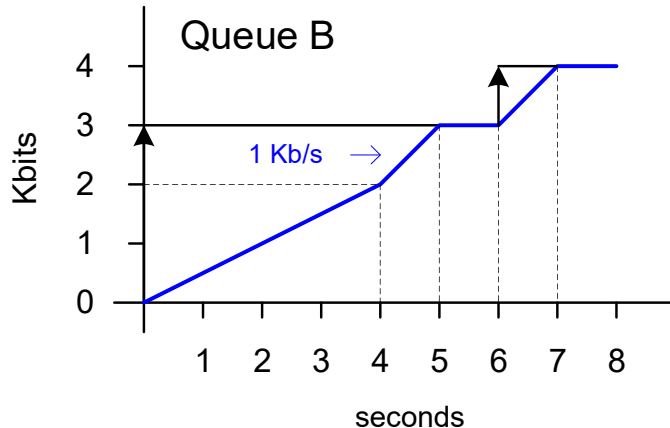
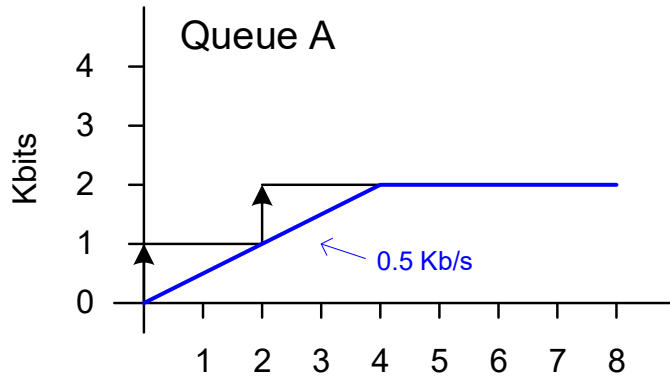
GPS scheduling



Packet scheduling



Weighted Fair Queuing (WFQ)



- In WFQ packets are served by the order in which they terminate service in the corresponding GPS system (if already arrived)

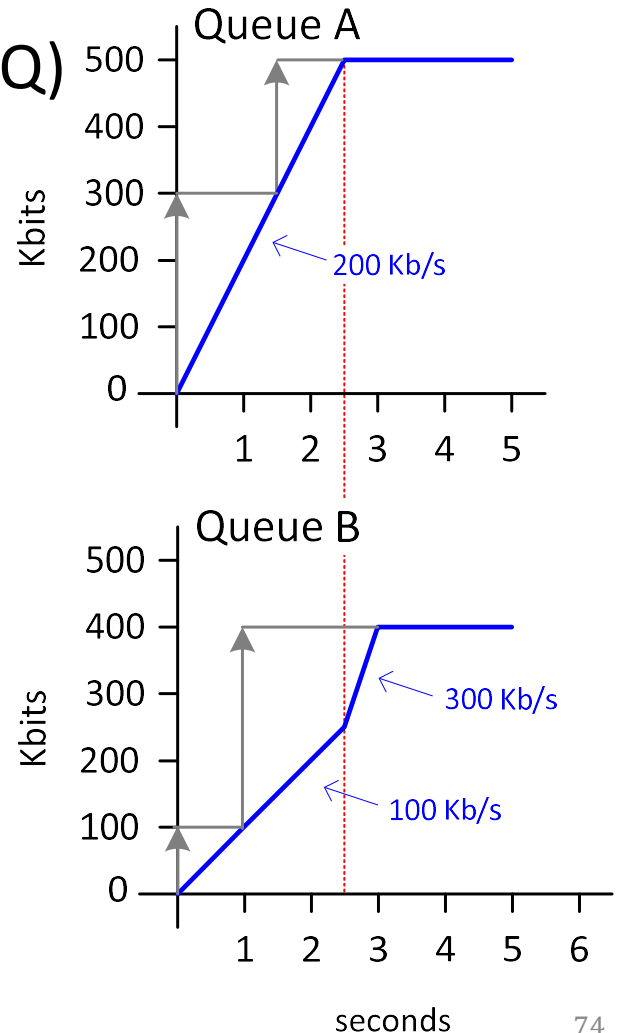
Example:

- $C = 1 \text{ Kb/s}$, equal weight queues
- Packets:
 - A1 (1 Kb, $t=0$), A2 (1 Kb, $t=2$)
 - B1 (3 Kb, $t=0$), B2 (1 Kb, $t=6$)
- Order in GPS:
 - A1 ($t=2$), A2 ($t=4$), B1 ($t=5$), B2 ($t=7$)
- Order in WFQ:
 - A1 ($t=1$), B1 ($t=4$), A2 ($t=5$), B2 ($t=7$)

Weighted Fair Queuing (WFQ)

Example:

- $C = 300 \text{ Kb/s}$, weight of queue A = 2, weight of queue B = 1
- Packets:
 - A1 (300 Kb, $t=0$), A2 (200 Kb, $t=0$)
 - B1 (100 Kb, $t=0$), B2 (300 Kb, $t=0$)
- Order in GPS:
 - B1 ($t=1$), A1 ($t=1.5$), A2 ($t=2.5$), B2 ($t=3$)
- Order in WFQ:
 - B1 ($t=1/3$), A1 ($t=4/3$), A2 ($t=2$), B2 ($t=3$)



Round Number (equal weights)

- Round Number (RN) – real variable that increases in time at a rate inversely proportional to the number of active ~~sessions~~ *queues* (N_{active}) during a time interval $[\tau_i, \tau_{i+1}]$ where the number of active ~~flows~~ *queues* is kept constant:

$$RN(\tau_i + t) = RN(\tau_i) + \frac{1}{N_{active}} t$$

- RN is reset to zero whenever the system gets empty
- The Finish Number (FN) of packet k with size L_k that arrives at queue i is given by:

- If the queue has packets waiting,

$$FN_{i,k} = FN_{i,k-1} + \frac{L_k}{C}$$

- If the queue is empty,

$$FN_{i,k} = RN + \frac{L_k}{C}$$

Round Number (equal weights)

- In the GPS system, a packet finishes service when $RN = FN$
- Once assigned the FN does not depend on future arrivals or departures
- The time instant when packets finish service in the WFQ system does not coincide with the FN
- WFQ serves packets according to their FN

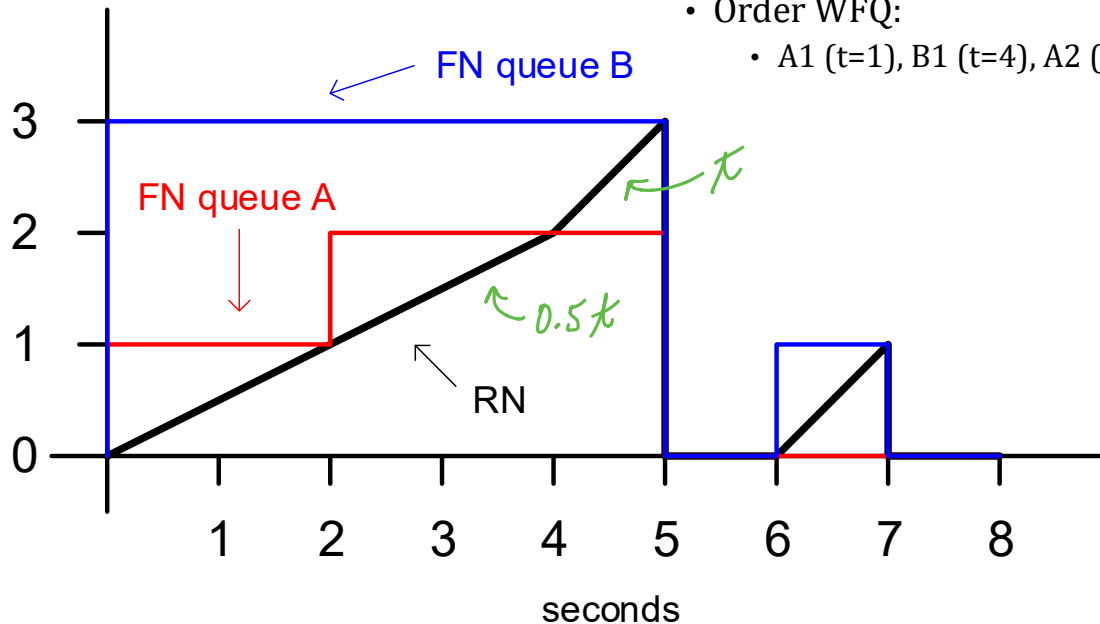
Example with RN

or

$$FN_k = FN_{k-1} + L_k / C$$

$$FN_k = RN + L_k / C$$

- $C = 1 \text{ Kb/s}$, queues with equal weights
- Packets:
 - A1 (1 Kb, $t=0$), A2 (1 Kb, $t=2$)
 - B1 (3 Kb, $t=0$), B2 (1 Kb, $t=6$)
- Order GPS:
 - $FNA1 = 1$ ($t=2$)
 - $FNA2 = 2$ ($t=4$)
 - $FNB1 = 3$ ($t=5$)
 - $FNB2 = 1$ ($t=7$)
- Order WFQ:
 - A1 ($t=1$), B1 ($t=4$), A2 ($t=5$), B2 ($t=7$)



Round Number (different weights)

- During a time interval $[\tau_i, \tau_{i+1}]$ where the number of active sessions is kept constant,

$$RN(\tau_i + t) = RN(\tau_i) + \frac{1}{\sum_{j \text{ active}} \phi_j} \quad \text{✗}$$

- RN is reset to zero when the system gets empty
- The Finish Number (FN) of packet k with size L_k that arrives at queue i is given by:

- If the queue has packets waiting,

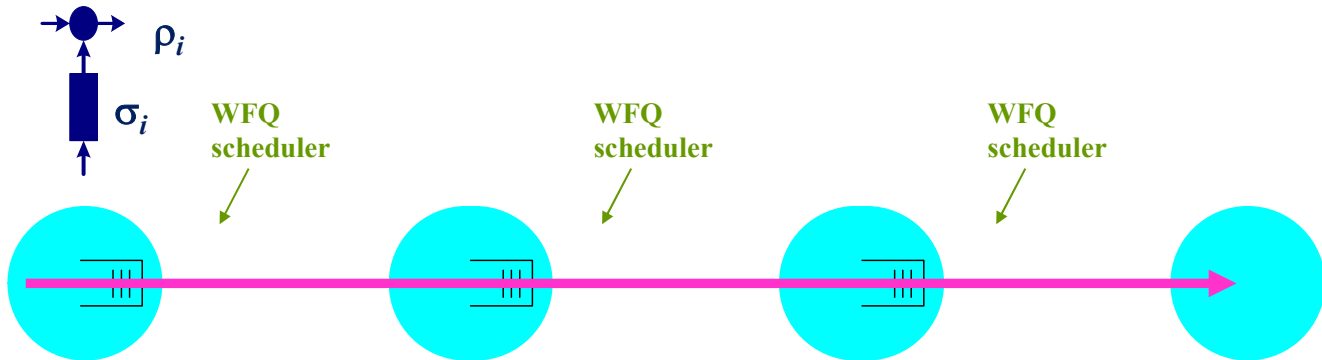
$$FN_{i,k} = FN_{i,k-1} + \frac{L_k}{C\phi_i}$$

- If the queue is empty,

$$FN_{i,k} = RN + \frac{L_k}{C\phi_i}$$

Deterministic QoS guarantees: an important formula

leaky-bucket



Maximum end-to-end delay of session i , for m hops:

$$D_i = \frac{\sigma_i + (m-1)L_{\max}}{\rho_i} + \sum_{j=1}^m \frac{L_{\max}}{C_j}$$

A. Parekh, R. Gallager "A generalized processor sharing approach to flow control in integrated services networks: the multiple node case", TON, 1994